SolariNet — Directory Services & SSO

A cross-platform identity backbone for the homelab: Active Directory, Kerberos, LDAP, and web SSO — built on `radium` · 2026-07-05

> **What went live.** `radium` (a Pi CM5, reinstalled to Debian 13 on NVMe) is now a **Samba Active Directory domain controller** for `AKORIA.ORG`, with **Keycloak** layered on top for web/OIDC single-sign-on federated to AD, a first fleet host (`tungsten`) joined and proven end-to-end, an NFS home export live on `cesium`, three new Ansible roles, and a complete (undeployed) OIDC integration for the SolariNet dashboard.

Architecture

One directory, every platform. Samba AD DC is the authoritative identity store (Kerberos + LDAP + DNS). Linux fleet hosts join it via SSSD/realmd; macOS/Windows can join natively; websites federate through Keycloak (OIDC/SAML). Home directories live once on `cesium` and follow the user to any host via autofs+NFS. UIDs are SID-algorithmic, so a user is the same numeric identity fleet-wide.

```
mac / windows / linux / ssh ──▶  Samba AD DC (radium.akoria.org)  ◀── SSSD/realmd (fleet
                                   ├─ Kerberos KDC (:88)   ├─ LDAP/LDAPS (:389/:636)
websites / dashboard ──OIDC──▶  Keycloak (sso.akoria.org:8443) ──LDAPS──▶ AD federation
any joined host ──login──▶  autofs /home/akoria/<user>  ──NFS──▶  cesium:/export/home/<u
```

What's live

COMPONENT	WHERE	STATE
Samba AD DC — `AKORIA.ORG`	radium 10.1.0.10	live · Kerberos/LDAP/DNS/SMB verified

Keycloak 26.6.4 (PostgreSQL, HTTPS :8443)	radium · sso.akoria.org	live · realm `akoria`, AD federated over LDAPS
OIDC client `solarinet-dashboard`	Keycloak akoria realm	created · AD login → token verified
Fleet join proof	tungsten 10.5.0.5	joined · SSO login + auto-home + TGT verified
NFS home export	cesium `/export/home`	live · root_squash, crossmnt
Ansible roles: directory-join / shared-home / dev-tools	deploy/fleet/ansible	authored + validated (check-mode) · uncommitted
Dashboard OIDC login	dashboard/API candidate	locally tested candidate · unpublished and not deployed

Credentials (AD Administrator, Keycloak admin, DB, dashboard client secret, AD user `jason`) are stored in agent memory and in `/root/.kc-*` on radium — deliberately kept out of this committed doc.

Ansible (deploy/fleet/ansible)

Three opt-in roles were added alongside the existing base-os/common-stack/solarinet-client, wired into `site.yml` and gated by `solari_ad_enabled` / `solari_home_enabled` / `solari_devtools_enabled`, with the AD contract in `group_vars/all.yml`:

- **directory-join** — install SSSD/realmd stack, point DNS at the DC, idempotent `realm join`, declarative `sssd.conf` (SID id-mapping for stable UIDs), mkhomedir, AD-group sudo. Join password is supplied at run time via vault/`--extra-vars` and never stored.
- **shared-home** — autofs + NFS client; wildcard map mounts `/home/akoria/<user>` from cesium on demand.
- **dev-tools** — the CLI toolset (zsh, jq, cloc, bat, ripgrep, eza, fd, cloc, starship, thefuck…), system-wide oh-my-zsh + powerlevel10k, and the fleet-adapted dotfiles vendored from xenon (deployed via `/etc/skel` so new directory-user homes inherit them).

```
# run a role against a host (from benzene or xenon):
ansible-playbook -i inventory playbooks/site.yml --limit <host> --tags ad \
  -e solari_ad_enabled=true -e solari_ad_join_password='<AD admin pw>'
```

> **Dashboard OIDC — candidate gate.** The current SolariNet worktree contains the implementation and local test coverage, but it is *not published or deployed*. Local password login is a separate, disabled-by-default break-glass path. OIDC sessions recheck group membership through UserInfo every 60 seconds, bind the returned subject to the original identity, refresh expiring access tokens, downgrade to viewer on provider errors, and revoke on subject mismatch. Discovery metadata must identify the configured issuer; authorization, token, JWKS, and UserInfo endpoints are limited to HTTPS on the issuer origin or an explicitly configured `allowedEndpointOrigins` origin. Provider responses are capped at 1 MiB. Before it goes live:
>
> 1. Add the dashboard's public callback (`https://<dashboard-host>/api/auth/oidc/callback`) as a Valid Redirect URI on the `solarinet-dashboard` Keycloak client; add a *Group Membership* mapper (claim `groups`) so `domain admins` maps to the admin role.
> 2. In `solari-auth.json` → `oidc`: set `enabled:true`, `redirectUri`, and `caBundle` to a PEM trusting radium's cert (`openssl s_client -connect sso.akoria.org:8443 … > radium-ca.pem`). Configure `allowedEndpointOrigins` only if the provider uses additional HTTPS origins; entries must be origins such as `https://sso.akoria.org:8443`, with no path.
> 3. Ensure the dashboard host resolves/reaches `sso.akoria.org:8443`; deploy; browser-test the "Sign in with Akoria SSO" button.

Deferred (your call)

- Commit the Ansible roles + dashboard OIDC work (currently in the working tree).

- Roll out dev-tools/shared-home to real hosts; live end-to-end shared-home test (login → NFS home).
- Join production hosts — xenon (⚠ the box the agent runs on; keep a fallback login), cesium, benzene.
- DNS reach for non-fleet mac/Windows clients so they can find the DC.
- Replace Keycloak's self-signed cert + Samba's autogenerated TLS cert with a proper CA/SAN cert.
- Kerberized NFS (sec=krb5p) to replace sec=sys for the home export.